

Embracing Tradition Through Technology: The Mixed Reality Calligraphy Studying Environment

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ABSTRACT

This article introduces an innovative Mixed Reality (MR) system designed explicitly for calligraphy learning and practice. Learners must prepare numerous copies of calligraphy works and character templates in traditional calligraphy study and practice. They often rely on oral guidance from teachers in person for their calligraphy practice. However, with the progress of digital technology, we envision leveraging MR wearable glasses combined with image capture and analysis techniques to assist calligraphy learners in improving their practice in a more flexible time.

Index Terms: Mixed reality—Image analysis—Human-centered computing—Human-computer interaction—Immersive study;

1 INTRODUCTION

Traditional calligraphy, entrenched in the master-apprentice dynamic, demands a significant investment of time and energy from both teachers and students [3, 4, 6]. To enhance the learning experience, our project incorporates MR Technology, creating an immersive calligraphy studying system. Our design, the Immersive Calligraphy Studying System, allows users to operate in a visible, immersive environment [2, 5, 8], and offers real-time feedback and intuitive interaction with virtual elements. This approach provides a more natural experience than conventional AR or VR systems, such as the VR project 'You, Calligrapher' by Patha Pradipa, which relies on VR handles for simulating calligraphy [1]. Our design's key innovation is the unique integration of MR technology in calligraphy training, providing real-time instruction and practice feedback, thus effectively bridging the gap between traditional methods and modern digital practices. We aim to evolve MR-assisted learning by combining ongoing research with experimental design, ultimately leading to the completion of a comprehensive and immersive calligraphy study system.

2 DESIGN CONCEPTION

We initiated research among various calligraphy study groups to create an MR system that enhances calligraphy learners' practice and learning experience. At this stage, we intend to recruit volunteers extensively to enrich our research content, make further references, and for subsequent experiments. Our ongoing research comprises two main steps.

First, we ask volunteers to use a calligraphy brush to write certain characters and record the writing process. This step aims to collect data from as many calligraphy learners as possible, capturing the dynamics of their hand movements and the changes in their writing styles during the process. These data will serve as a valuable reference for further program development. Another step is to gather these volunteers' feedback and evaluate their feelings and further insights into their ideal study requirements. We are continuously working on this process to collect as much user feedback as possible

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Figure 1: The 'Immersive Calligraphy Studying System' usage diagram is presented. We intend to utilize Unity with Mixed Reality technology to achieve educational and interactive functionalities, aiming to assist calligraphy learners in improving their practice and providing an immersive studying experience. In this picture, 'p1-p9' represents different interactive stages. Credit goes to the author.

for further evaluation. The table below (Figure 4) is our evaluation of the user questionnaire to realize potential users' attitudes. For the **MR immersive calligraphy studying the system**, we have the following conceptions:

1. **Interactive:** Taking Mixed Reality glasses, for example, 'Microsoft HoloLens' as a carrier lets users wear (Figure 3).
2. **Image data analysis and storage:** Based on Unity to develop an interactive system, as Figure 1, p1-p3 shows, user can reach out and grab the floating green blocks in front of their eyes. Each block represents different calligraphic characters, and users can choose the fonts they want to practice according to their requirements. For this step, we intend to take as many Traditional Calligraphy Font Copying Materials to integrate and select more representative fonts such as *Zhao, Liu, Ou, and Yan style characters (four great master characters of ancient Chinese calligraphy)* [7] to collect more data and integrate more data. Here, 'graphic data' refers to images of various fonts, which we intend to store within our system. These images will be used as a reference when users view them through glasses and practice calligraphy. The system will analyze and compare the font that users write with the original image.
3. **The system gives instructions:** Practice copying the font while the user can see the font through the MR glasses (Figure 1, p4-p6). This practice system will give the user writing instructions by taking the user's hand and the font's shape simply by writing it down. For this step, we plan to integrate external computer assistance to analyze the character images users write, aiming to enhance analysis efficiency and improve the overall experience for users.
4. **The system gives feedback:** Considering that in traditional calligraphy studying, the verbal and practical transmission between master and apprentice is an integral part of the learning

$$\text{Percentage}_{\text{Option 1-5}} = \left(\frac{\text{Number of People Choosing the Option}}{\text{Total Number of Respondents}} \right) \times 100$$

Figure 2: Statistical formulas for questionnaire feedback, used for a more intuitive visualization of user attitude.

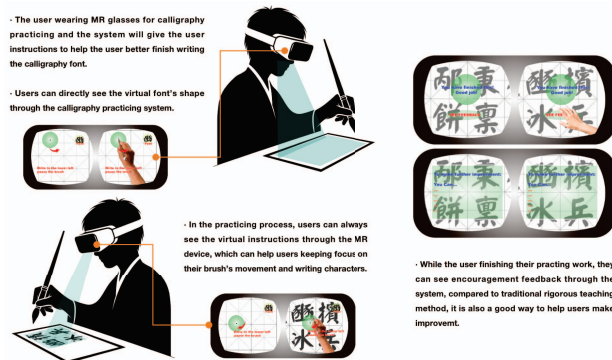


Figure 3: User wearing MR glasses for calligraphy practicing. Credit goes to the author.

calligraphy process, we plan to add a feedback mechanism. After finishing writing, users can see their practice feedback from the MR glasses (Figure 1, p7-p9).

By using MR facilities combined with Unity interactive technology, calligraphy learners can practice writing characters more flexibly and gain corresponding instructions and feedback without the guidance of a teacher nearby.

3 PROJECT EVALUATION

We have now outlined four feasible concepts, as mentioned above. For this project, we intend to utilize Mixed Reality technology to provide calligraphy learners with an immersive learning environment. Compared to calligraphy learners who practice in isolation without the guidance of a nearby teacher, proven assisted studying facility Mixed Reality glasses (e.g., 'Microsoft HoloLens') combined with our system can help calligraphy learners enhance their sense of engagement and concentration during practice.

In contrast to forepassed VR- or AR-based projects like Patha Pradipa's project 'You Calligrapher' [1], our design can overcome the limitations of VR facilities, which have a limited visual range and AR technology that cannot create an immersive experience environment similar to VR. Our design provides users with a visible virtual practicing environment through MR-based technology (Figure 3). By giving users instructions in real-time and offering positive feedback after users practice, our design mitigates 'the pressure of unwavering obedience and reverence towards the teacher within the framework of Confucian education' [6]. To create a more relaxed learning atmosphere for users.

4 FUTURE WORKS AND APPLIACATION

To further improve this project, we will make the following perfections: **Expanding the content and styles** in the system, continuously adding more calligraphic styles and fonts to cater to a wider range of users and preferences, and working with calligraphy experts to expand additional valuable and professional suggestions. Furthermore, to **enhance the interactivity of the**, we explore the gesture and graphic recognition part more, improving the system efficiency in capturing written font's shape-changing and writing posture (e.g., hand & calligraphy brush part) for more natural interactions. Meanwhile, we will invite more volunteers in subsequent experiments to understand user evaluations and feelings towards the

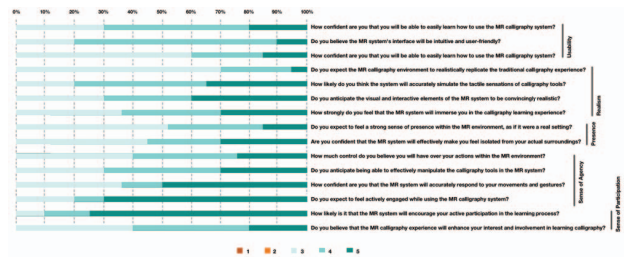


Figure 4: 'Participant Response Distribution for MR Calligraphy System Evaluation.' By convening volunteers (including 20 females and one teacher Agemean=21 and 20 males except for one teacher Agemean=22.5), who have varying experience in calligraphy learning to conduct preliminary research work and finish a questionnaire, we conducted the quantifiable table above. This table reflects participants' evaluation of our current investigation and work, including participants' ratings on usability, realism, presence, sense of agency, and sense of participation (for option 1-5: 1= Strongly disagree, 5=Strongly agree). We use a formula to analyze the feedback from the questionnaire (Figure 2). Credit goes to the author.

system feedback mechanisms and to see how users will evaluate their calligraphy practice feedback. Therefore, we can, based on the feedback of the volunteers, make more feasible **technical improvements and perform iterations** to make the interaction and the feedback mechanism more intuitive and understandable.

5 CONCLUSION

This article introduced a Mixed Reality system for calligraphy learning, combining traditional art with modern technology to create an immersive, interactive experience. Our approach overcomes the limitations of conventional learning methods, providing real-time guidance and feedback, thus enhancing the efficiency and enjoyment of learning calligraphy. The balance between maintaining the authenticity of calligraphy and utilizing digital advancements is critical to our design philosophy. Future enhancements will focus on expanding the system's capabilities and exploring its application in other artistic domains. Our project combines art education and technical operation, bridging the gap between ancient practices and the digital era.

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